Document Identifier: **PMC8195256**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: P2012).
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
We put three replicates of three independent data in Graphpad software.
Number: We put three replicates of three independent data in Graphpad software.
Cell Line Authentication
not detected.

Data Information	
Identifiers: Analysis of mtDNA damage by quantitative PCR (qPCR) mtDNA damage was analyzed by qPCR, as described previously 56,57,70.	56,57,70

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Anti-APE1 (ab194), anti-PRMT1 (ab12189), anti-GAPDH (ab9485), anti-COX IV (ab16056), anti-TOM20 (ab56783), anti-Mia40 (ab87033), anti-mono methyl arginine and dimethyl arginine (ab412), and anti-b tubulin (ab6160) antibodies were from Abcam (Cambridge, UK) the anti-Histone H3 antibody was from Cell Signaling Technology (Danvers, MA, USA 4499) anti-b-actin (GTX11003) anti-GFP antibodies were from GeneTex (Irvine, CA, USA 26673) and anti-asymmetric dimethyl-arginine (ASYM24, 07-414), anti-symmetric dimethyl-arginine (SYM10, 07-412), and anti-mitochondrial HSP70 antibodies were from Upstate (Lake Placid, NY, USA H1830-91A).	anti-PRMT1		Suggestion: (DSHB Cat# PCRP-PRMT6-1H4, RRID:AB_2619009)(link)
	anti-GAPDH		
	anti-COX		Suggestion: (Karen Menuz, University of Connecticut Cat# anti-Amt, anti-CG6499, RRID:AB_2864754)(link)
	anti-TOM20		
	anti-Mia40		
	anti-b-actin		Suggestion: (MBL International Cat# PM053-7, RRID:AB_10697035)(link)
	anti-GFP		
Immunoprecipitation using APE1 antibody was used to enrich APE1 to detect its methylation status.	APE1		
Experimental Models: Cell Lines			
Plasmids and recombinant proteins Human APE1 sequence was PCR amplified from HEK293 cell cDNA.	HEK293		Suggestion: RRID:CVCL_0045(link)
SgRNA-plasmids were premixed (1:1:1) and cotransfected into HEK-293 cells with the donor, using Lipofectamine 2000 (Life Technologies, Carlsbad, CA, USA, 1 mg: 0.1 nmol).	HEK-293		Suggestion: RRID:CVCL_0045(link)
HEK-293-R301K cell lines were used in APE1 mitochondrial localization experiments and cell survival assays after H2O2 treatment.	HEK-293-R301K		
Briefly, HeLa cells were washed twice with 1xPBS and resuspended in 500 ml buffer A (10 mM Hepes, 10 mM KCl, 0.1 mM MgCl2, 0.1 mM EDTA, 0.1 mM DTT, 5 mM PMSF, pH = 7.9).	HeLa		Suggestion: RRID:CVCL_0030(link)
HEK293 cells and HEK293-APE1-R301K KI cells were used in this assay.	HEK293-APE1-R301K		
Recombinant DNA			

His-tagged APE1 was cloned to pET-14b plasmid.	pET-14b		
APE1 mutant plasmid were PCR generated using wild type pET-14b-His-APE1 as template with TianGen Fast HiFidelity PCR Kit (KP202) following standard protocol.	pET-14b-His-APE1		
The pCDNA6.1-APE1-R301F plasmid was transfected into R301K KI cells and labeled R301F group.	pCDNA6.1-APE1-R301F		
Cells were transfected with APE1-WT, APE1-R301K, or APE1-R301F plasmids using Lipofectamine 2000.	APE1-R301F		
MTT cell survival assay Cells were transfected with pCDNA6.1-APE1-WT, APE1-R301K, or APE1-R301F plasmids using Lipofectamine 2000.	pCDNA6.1-APE1-WT		
Software and Algorithms			
Images were collected and processed using Zeiss blue software and sized in Adobe Photoshop CS5.0.	Adobe Photoshop		Suggestion: Adobe Photoshop, RRID:SCR_014199(link)
We put three replicates of three independent data in Graphpad software.	Graphpad		Suggestion: GraphPad (RRID:SCR_000306)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Statistical analysis We used students t-test for all comparisons between untreated and treated groups.	t-test
Data passed DAgostino-Pearson omnibus normality test and F test.	Data passed DAgostino-Pearson omnibus normality test
	F test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq